

Multimodal Disease Classification Using Chest X-Rays and Clinical Reports: A Mixture-of-Experts Fusion Approach

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Abstract—Identifying thoracic illnesses using chest x-rays can only occur through utilizing both visible and written clinical information which is how radiologists employ their workflows. A new methodology, the multi-modality mixture-of-experts (MoME) model, sees use with multi-label thorax disease diagnosis for the purpose of using chest x-ray images and clinical documentation in combination. The methodology incorporates anterior-posterior (AP), posterior-anterior (PA), and lateral chest x-ray views, paired with results and impressions in reports in order to increase diagnostic accuracy. DenseNet-121 separate expert encoders are developed by view type,—as represented by the three aforementioned views,—while the inadequacy of clinical text representation is addressed by PubMedBERT. Each expert encoder is connected to an evaluate-to-gate (E2G) network which produces a unique set of modality-specific weights that allow for adaptive multi-modality fusion based on sample identification. Experiments utilize the MIMIC-CXRS dataset for diagnosis of 14 thoracic diseases using two performance metrics: the macro area under the receiver operating characteristic curve (AUROC) and the macro F1 score. The proposed schema’s macro F1 score equals or exceeds the base maximum by approximately 8% and produces interpretable sample-by-sample modality attribution via the evaluate to gate network. Class imbalance is mitigated using class weighted binary cross-entropy loss during training. The proposed schema establishes a validated reproducible framework for multimodal thoracic diseased classification within vision to language healthcare AI systems.

Index Terms—multimodal learning; chest X-ray; clinical NLP; DenseNet; PubMedBERT; mixture of modality experts; disease classification; MIMIC-CXR ; MIMIC-IV; EHR ; Clinical Notes; Fusion modal

I. INTRODUCTION

Pulmonary and cardiac diseases are two most common causes of morbidity in the population, which require accurate and timely radiological diagnosis. The most commonly used diagnostic imaging study in the world is a Chest X-ray (CXR) but its interpretation is only possible when both visual findings and the history of the patient are considered concomitantly in radiology reports, when creating the message used in medical imaging studies. Over this clinical fact, most automated classification systems work on only one modality, either images or text and do not utilize the complementary information presented in the other. Convolutional

neural networks like DenseNet-121 have enabled remarkable achievements in image-based CXR analysis, under specific conditions, matching the performance of radiologists [1]. At the same time, transformer-based language models, which was pretrained on biomedical corpora, including BioClinicalBERT [2] and PubMedBERT [3] have significantly enhanced clinical natural language processing (NLP). Nevertheless, systems merging both modalities in an end-to-end trainable manner are under-investigated, especially in the case of multi-label, multi-view model. The following are key challenges associated with

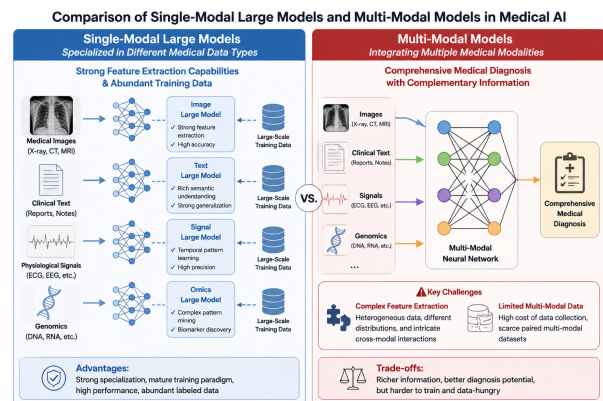


Fig. 1. Evaluation of the Comparison Unimodal Large Models versus Multimodal Models in terms of Medical AI. Left: single modal large models designed to store many different types of medical data, with a robust feature extraction abilities, as well as an abundance of training data. Right: Multimodal neural network model designed to utilize multiple modalities of medical data to provide comprehensive medical diagnoses with issues surrounding complex feature extraction approaches and limited availability of multimodal medical data. The background’s color differences (blue versus red) represent a fundamental trade-off in these two approaches.

multimodal CXR classification: (i) the heterogeneity between image tensors and text sequences of tokenized text does not accommodate a notion of naive early fusion; (ii) possibly variable modality availability at inference time (not all patients will provide lateral view or complete an entire sequence of tokenized text); (iii) an extreme imbalance of classes, as 14 pathology labels are defined by the CheXpert labeling

convention; and (iv) the need to make interpretable predictions that can be used in clinical practice. We deal with these issues using MoME, a Mixture-of-Modality Experts network, which contains separate, view-specific DenseNet-121 image encoders and PubMedBERT text encoder. The gating network is trained to dynamically weight each expert, which produces a meaning in which the contribution of each of the different modalities can be given to the final prediction. The design facilitates smooth degradation in cases where modalities are not present and gives the clinicians a clear signal of which inputs may have influenced the classification. The rest of this paper is divided in the following way. In Section II, related literature on multimodal medical image classification is surveyed. The dataset and preprocessing pipeline are described in section III. Section IV describes the architecture and training plan of MoME. Experiment results and ablation analyses are reported in Section V. Section VI addresses the findings, limitations, and future directions and section VII is the conclusion.

II. LITERATURE SURVEY

A. Unimodal Chest X-Ray Classification

By utilizing DenseNet-121, trained using the ChestX-ray14 data set, CheXNet was able to detect pneumonia with a performance level greater than that produced by practising radiologists. This methodology was then extended to include 14 pathology labels with the use of uncertainty labeling and was established as a standard benchmark for subsequent works, including the CheXpert data set. Nonetheless, these image-only detectors provide a strong baseline against which further work can be conducted, but do not consider textual context that radiologists routinely use in diagnosing a patient.

B. Clinical NLP And Text Based Diagnosis

The BERT model, improved by Bio_ClinicalBERT, has been fine-tuned on MIMIC-III discharge notes (National Institutes of Health, Libr.), providing better representations for clinical NLP tasks (National Institutes of Health, Libr.). Additionally, PubMedBERT, trained specifically on biomedical literature, provides better embeddings than the models above for medical terms. However, the above models do not provide sufficient understanding for visual representation and diagnostic support when used on their own in imaging diagnosis tasks involving chest X-ray (CXR) diagnostic tasks.

C. Multimodal Fusion in Medical Imaging

Medically-oriented classifications are a particularly interesting and dynamic area of research and development. Li et al. [5] evaluated 114 candidate publications on deep learning-based multimodal fusion methods and constructed an overall taxonomy of methods for classifying images from different imaging modalities such as brain, lung, and cardiac. Through their review of the literature, Li et al. identified five key paradigms for multimodal fusion: input, output, values of single-level intermediate, hierarchical attention, and output values. Notably, single-level intermediate fusion methods feature concatenation of the outputs from two (or more) encoder

branches prior to the final classification of an image, and this method has been employed in 51% of the studies examined by Li et al.. Li et al. found that single-level intermediate fusion consistently resulted in improved classification accuracy when compared to both input- and output-based fusion methods across heterogeneous modality pairings. By comparing their results using multimodal fusion versus the best classification performance reported by single-modality classifiers on the ADNI, ABIDE, and GAMMA datasets, Li et al. achieved an increase in classification performance of between 8 and 17 percentage points.

ConVIRT [6] is the first method of using visual-language pretraining on chest X-ray images and their corresponding radiology reports using contrastive learning objectives in the process. The resulting performance demonstrated strong capabilities for few-shot classification with respect to images and radiology reports. Additionally, Phrase-grounded pre-training was created as an extension of the previously developed multi-label classification paradigm described by BioViL [7]. However, it should be noted that these are still not end-to-end discriminative training-based representations for multi-label classification purposes.

The OpenI dataset [8] from Indiana University consists of 3,955 radiology reports and 7,470 chest X-ray images that were previously used to provide an early benchmark in the multimodal radiology AI field. More recent publications have used MIMIC-CXR [9], which includes over 220,000 chest radiographs with associated radiology reports from the Beth Israel Deaconess Medical Center, to enable the training of more generalizable models. There are four main areas where our research contributes to this body of knowledge: (i) our research uses the larger MIMIC-CXR dataset compared to OpenI; (ii) we use three different views (AP, PA, and lateral) of the chest simultaneously; (iii) we developed an adaptive gating mechanism that allows for modality attribution; and (iv) our framework can be trained end-to-end and generates clinical interpretability using Grad-CAM visualizations.

III. DATASETS AND PREPROCESSING

A. Dataset and Ethical Access

This research project utilizes 2 datasets available through the PhysioNet platform (PhysioNet.org), which are provided by the Medical Information Mart for Intensive Care (MIMIC) [13]. All authors have completed the Collaborative Institutional Training Initiative (CITI) "Data or Specimens Only Research" certification and completed an individual PhysioNet Data Use Agreement before accessing the dataset. Because the MIMIC data are fully de-identified in accordance with HIPAA Safe Harbor standards, there was no requirement for separate institutional review board exemption prior to using the dataset.

Data used in the analysis include chest x-rays from the MIMIC-CXR dataset ($n \approx 220,000$) collected from patients ($n \approx 65,000$) who were admitted to the Beth Israel Deaconess Medical Center. Each chest x-ray may contain up to 3 view types (posteroanterior, anteroposterior, and lateral); in addition, CheXpert's automated labeling pipeline [4] generates

disease labels for 14 pathology categories (atelectasis, cardiomegaly, consolidation, edema, enlarged cardiomeastinum, fracture, lung lesional opacity, no finding, pleural effusion, pleural other, pneumonia, pneumothorax, support devices) and uses the uncertain-as-positive convention for multi-label classification.

TABLE I
LABEL PREVALENCE ACROSS THE DATASET SAMPLE ($N = 220,183$)

Condition	Count (N^+)	Prevalence (%)
Atelectasis	54,414	24.7%
Cardiomegaly	49,058	22.3%
Consolidation	14,576	6.6%
Edema	38,751	17.6%
Enlarged Cardiomeastinum	15,884	7.2%
Fracture	4,763	2.2%
Lung Lesion	7,108	3.2%
Lung Opacity	53,407	24.3%
No Finding	73,112	33.2%
Pleural Effusion	58,151	26.4%
Pleural Other	2,637	1.2%
Pneumonia	33,597	15.3%
Pneumothorax	11,248	5.1%
Support Devices	64,829	29.4%

The radiology report text comes from MIMIC-IV [14], the latest version of MIMIC’s clinical database which contains free-text reports associated with MIMIC-CXR’s patients and studies. MIMIC-IV report text has much better delineation of sections than does the report text in MIMIC-CXR and contains a wider date range of reports. Reports were paired with their imaging study due to matching on both subject identifier and study identifier, resulting in a complete paired image–text corpus.

B. Data Preprocessing and Text Extraction

Through the use of regular expressions, radiology reports from MIMIC-IV can be parsed into three clinically important sections (i.e., indication – the indication records patient history, an examination clinical indication; ii) findings – finds, and a systematic list of how each anatomical region was observed by the Radiologist; iii) Impression - summarises the main diagnosis). The boundaries of these sections were identified using keyword matching with header labels that are both standard and/or have variants (i.e., “Findings:,” “Finds:,” “Impression:,” “Results”). If there are multiple reports associated with any single study (e.g., addendum), the sections therein will be concatenated and separated with a token. The above three sections will then be constructed into a single structured input string:

INDICATION

FINDINGS

IMPRESSION

Valid and load images as per the view type. For example, if the only view that exists includes both AP and PA or lateral only, all imaging would be considered to have been

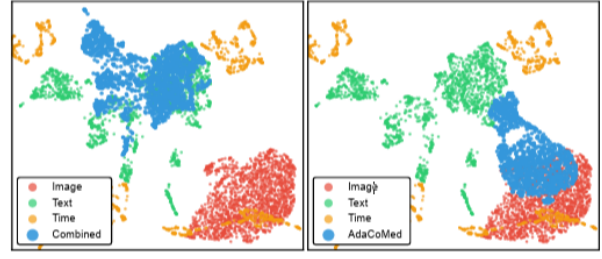


Fig. 2. The UMAP visualizations illustrate how the distribution of the features differs between the direct combination (left) and our proposed method (right). The direct combination feature distributions (blue) overlaps the text feature distributions (green), indicating a clear predominance of the text modality. Our model, on the other hand, is able to effectively integrate the modal clusters by bridging the distribution of features from image (red), text (green), and temporal (yellow) modalities.

performed on the segregent between AP and PA and therefore have insufficient primary reason/necessary bases for imaging. Likewise, for Text Quality assurance - if both the findings and impression sections are not available, all studies would not meet this standard and thus would be removed from consideration.

C. Data Splits

The dataset was separated into 3 splits: 70% for training, 15% for validation, and 10% used for testing. Each split of samples uses stratified sampling based on argmax value of the multi-label vector (proxy for primary class), thus maintaining approximate class proportions across splits. Summary statistics for the final dataset are displayed in Table I.

Partition	Count (N)	Percentage (%)
Train	165,166	75.0%
Validation	33,034	15.0%
Test	22,023	10.0%
Total	220,223	100.0%

IV. METHOD

A. Preliminary

The issue of mode dominance is one of the main problems associated with multimodal learning; an optimizer should ideally balance the gradients from the different modes so that each mode contributes to the final output equally, but this can result in the model relying on one mode significantly more than the others. In CXR classification, visual characteristics primarily drive the optimization process during the early phases of training. As a result, the contribution of the text branch, which contains clinically meaningful information, to the representation is limited; especially in diagnostically uncertain cases (e.g., subtle infiltrates or overlapping opacities) in which information found within the radiology report, specifically the impression and indication sections, offers important additional information. The imbalance associated with modality also appears when you consider the different views of the images. PA is the standard protocol for acquiring radiographs, and thus, is the most common view of the data. In comparison,

MoME: Mixture of Modality Experts for Multimodal Chest X-ray Report Classification

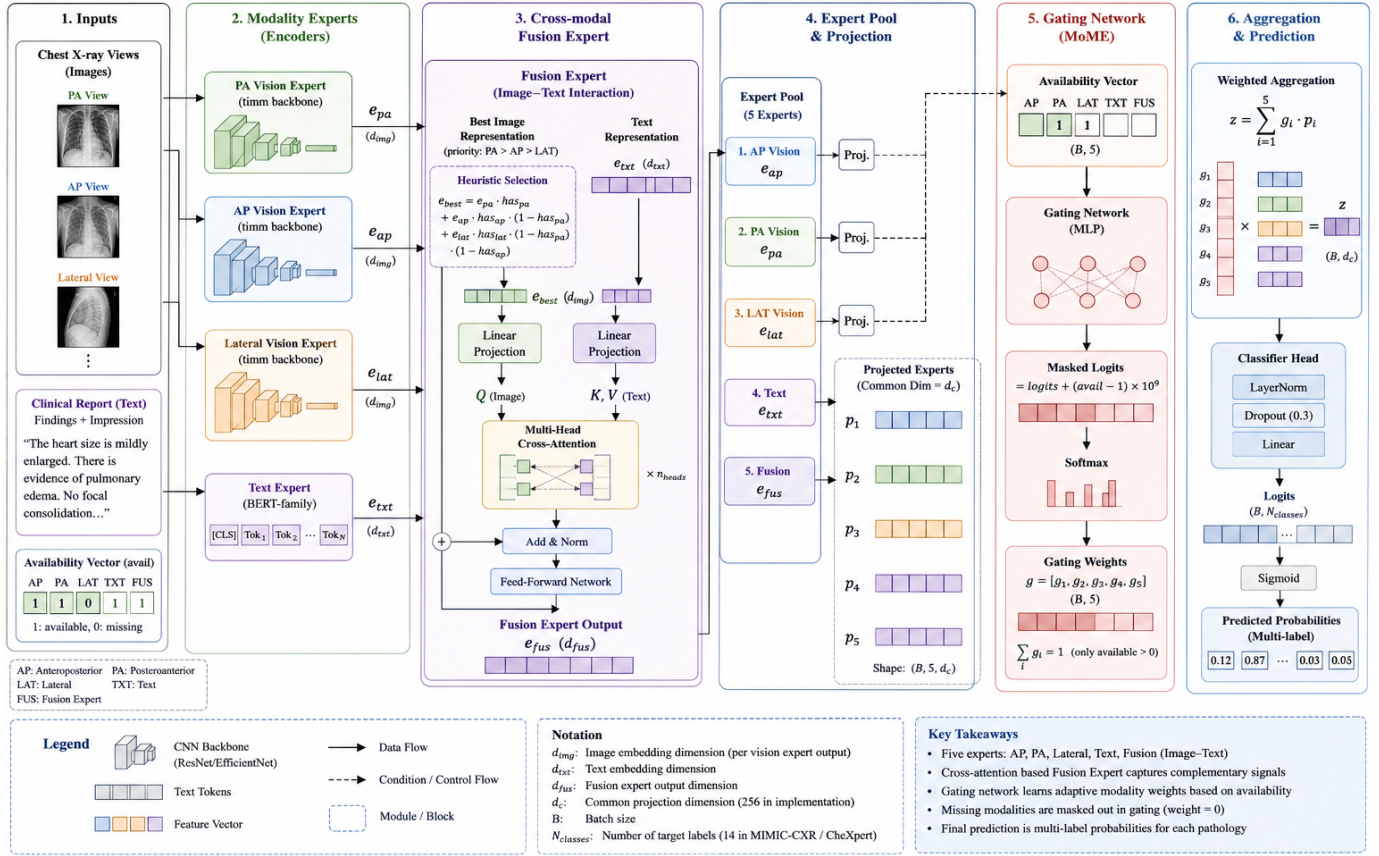


Fig. 3. Overall architecture of the MoME (Multimodal Expert) Classifier. The model integrates independent vision experts for AP, PA, and Lateral views with a BioBERT-based textual encoder. A dynamic gating network assigns importance weights based on modality availability vector $\mathbf{a}$ to produce the final fused representation.

AP and lateral views are much less common and are only acquired under special clinical circumstances. Most of the existing methods to combine the features from each of the views (e.g., concatenate features or average pool features) do so, assuming that the importance of the views is equal. This is not the case because only a small percentage of the available data consists of AP and lateral images, causing them to have much less of an effect on the model than they would otherwise. Therefore, the model is unable to take advantage of the complementary geometric and anatomical information provided by the multiple views.

A practical problem in multimodal clinical settings is missing modality inference. In practice, different modalities are not always available: many emergency or portable images are only available from the front view, and the text from a report may not be present at the time of inference. Multimodal systems that will function well across these types of submissions need to have a means of handling these types of situations in a manner that does not degrade the predictive performance of the model due to access to unreliable imputes or methods of failure. Therefore, it is important to design effective architectures that account for missing data input and meet the critical

requirement for clinical applications.

B. MoME: Addressing Domination with Expert Gating

We propose a Mixture-of-Modality Experts (MoME) approach to overcome the issue of dominating modalities, which occurs when modalities with significant relative density tend to dominate over other modalities. Rather than using a single method for combining modality feature vectors, we use dedicated expert encoders for each modality, with a centralized learned gating network that dynamically calculates the relative contribution of the various expert encoders based on the specific modality present for a given observation. This leads to three direct consequences. First, since each encoder's gradient is scaled relative to the respective gating weight, it effectively dampens the impact of the back-propagated gradient from any one over-represented stream, thereby preserving the ability to learn from the other, less represented streams of experts. Second, because the outputs of the gate softmax are constrained to sum to unity, this creates competitive conditions amongst the expert encoders such that no single modality can be monopolized. Third, by conditioning the gate to a binary availability vector, the model learns to reallocate gating weights

during inference for any modality that is missing—providing zero-shot missing-modality robustness without the need for additional imputation models or separate training regimes for missing modalities.

In practical terms, MoME consists of five experts. Each of the AP, PA, and lateral views has one expert that is a DenseNet-121 encoder. The combined indication-finding-impression text has one expert that is a PubMedBERT encoder. Finally, there is a fusion expert, which receives as input the mean pooled representation of all available image views, in order to capture cross-view consistency. A gating network, consisting of a single linear layer and applying a softmax function to a 5-bit availability vector is used to produce the expert weights for each sample in real time. The output of the gating network is then combined with the function output from the expert layers, producing a weighted sum

$$\left(z = \sum_{\tilde{w}} \alpha_{\tilde{w}} \cdot f_{\tilde{w}} \right)$$

that will be sent to the shared classification head. This architecture is described further in the following subsections.

C. Problem Formulation

In this study, we explore the possibility of multimodal classification of chest X-rays (CXRs) with multiple views and partial views of those CXRs being excluded from the output in professional radiology practice. A patient study may have several individual views (radiographic images) of the same examination (CXRs) and be accompanied by a free text report for each patient.

Formally, for each sample x , we define:

- **A set of radiographic images:**

$$I = \{I_{AP}, I_{PA}, I_{LAT}\}$$

- **A clinical report:**

$$T = \{w_1, w_2, \dots, w_n\}$$

- **A modality availability vector:**

$$\mathbf{a} \in \{0, 1\}^M, \quad M = 5$$

where each element of $\mathbf{a}$ indicates the presence of a corresponding expert input.

The objective is to predict a multi-label vector:

$$\mathbf{y} \in \{0, 1\}^C$$

where C denotes the number of clinical conditions (e.g., CheXpert labels).

D. Architecture

The Mixture of Modality Experts (MoME) framework integrates heterogeneous data types dynamically by using:

modality-specific encoders to encode each radiographic view and textual report separately; a cross-modal fusion expert that models the interaction between images and text; a gating network to dynamically weight the contribution of experts

based on the availability of the input data; an aggregate representation space that provides a shared location for all experts; and a multi-label classification head.

We model the two levels of data separately and the different types of data at each level without using traditional methods of fusing data (e.g. concatenating or averaging) and use a multi-level fusion approach to account for data that may be missing or have noise.

1) Vision Experts: Each of the three visual expert encoders consists of a **DenseNet-121** backbone, pre-trained on ImageNet. To adapt the architecture for feature extraction, we remove the final fully connected layer. The convolutional feature maps are processed via **Global Average Pooling (GAP)**, resulting in a 1024-dimensional vector representation:

$$\mathbf{v}_{img} \in \mathbb{R}^{1024}$$

All input images are resized to 224×224 pixels and normalized according to ImageNet statistics:

$$\mu = [0.485, 0.456, 0.406] \text{ and } \sigma = [0.229, 0.224, 0.225].$$

To enhance model robustness, we apply the following stochastic transformations during training:

- **Random Cropping:** Images are resized to 256×256 and randomly cropped back to 224×224 .
- **Geometric Transformations:** Horizontal flipping and random rotations within the range of $\pm 10^\circ$.
- **Color Jittering:** Adjustments to brightness and contrast by a factor of ± 0.2 .

For validation and testing, images are center-resized and normalized to ensure consistency.

To preserve the rich visual hierarchies learned during pre-training and prevent *catastrophic forgetting*, we employ a differential learning rate strategy. The backbone encoders are fine-tuned using a learning rate $\eta_{backbone}$ defined as:

$$\eta_{backbone} = \frac{1}{10} \eta_{new}$$

where η_{new} represents the learning rate assigned to the newly initialized layers.

Each radiographic view is processed by an independent **Vision Expert** encoder, enabling the architecture to capture view-specific anatomical features without interference. For a given image I_v , the corresponding embedding F_v is computed as:

$$F_v = \phi_v(I_v), \quad v \in \{AP, PA, LAT\} \quad (1)$$

where $\phi_v(\cdot)$ denotes a convolutional backbone (e.g., ResNet or EfficientNet via the `timm` library). To map raw features into a latent space suitable for multimodal fusion, the backbone is appended with:

- **Global Average Pooling (GAP)** to reduce spatial dimensions.

- **A Projection Head** consisting of the following sequence:

Linear $\rightarrow$ LayerNorm $\rightarrow$ GELU

The choice of independent visual experts is motivated by three primary factors:

- **Complementary Information:** Different radiographic views provide distinct perspectives; for instance, the lateral (LAT) view is often essential for resolving depth-related ambiguities present in frontal (AP/PA) views.
- **Feature Disentanglement:** Utilizing independent experts prevents feature entanglement, ensuring that the specific spatial characteristics of one view do not introduce noise into the representations of another.
- **Transfer Learning:** By utilizing backbones pre-trained on large-scale datasets, the model leverages robust visual priors, significantly accelerating convergence on medical-specific tasks.

2) *Text Expert:* The clinical text is encoded using PubMedBERT [3] (microsoft/BioMedNLP PubMedBERT-base-uncased-abstract-fulltext), which was pre-trained on abstracts, and full-text publication from Pubmed. The [CLS] token embedding from the final transformer layer provides a representation of the text in a 768-dimensional vector space. Text sequences were tokenized with a maximum length of 256 tokens using padding and/or truncation as necessary. During the training phase, all BERT parameters were fine-tuned with a reduced learning rate so as to retain the knowledge about the domain.

Given the tokenized input consisting of `input_ids` and an `attention_mask`, the global contextual representation F_T is obtained as:

$$F_T = \psi(T) = \text{Proj}(\text{BERT}_{CLS}(T)) \quad (2)$$

where:

- $\text{BERT}_{CLS}(T) \in \mathbb{R}^d$ represents the final hidden state of the [CLS] token, serving as the aggregate sequence representation.
- $\text{Proj}(\cdot)$ is a linear projection head that aligns the textual embedding dimensionality with the visual feature space to facilitate multimodal fusion.

The integration of textual features is based on the following considerations:

- **Semantic Depth:** Clinical text often contains nuanced diagnostic cues, such as patient history or subtle radiologic observations, that may not be readily apparent through visual features alone.
- **Global Summarization:** Utilizing the [CLS] token provides a condensed, global summary of the entire report, which is effective for high-level multi-label classification tasks.
- **Domain-Specific Priors:** By leveraging models like BioBERT, the system benefits from weights pre-trained on large-scale medical corpora (PubMed), ensuring the model understands specialized clinical terminology.

3) *Cross Modal Fusion Expert:* We develop a multi-head cross-attention-based Fusion Expert for an explicit interaction model for modalities.

The image selection strategy is built on the ability to construct a representative image embedding given that multiple views may exist:

$$F_I = \begin{cases} F_{PA}, & \text{if available} \\ F_{AP}, & \text{else if available} \\ F_{LAT}, & \text{otherwise} \end{cases}$$

4) *Cross Attention Mechanism:* The fusion specialist will calculate: To integrate the visual and textual representations, we use a cross-attention mechanism where the image features act as the query and the textual features provide us with the context. The Query (Q), Key (K), and Value (V) matrices are computed as:

$$Q = W_q F_I, \quad K = W_k F_T, \quad V = W_v F_T \quad (3)$$

The attention output is defined as:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) V \quad (4)$$

where d is the scaling factor corresponding to the embedding dimension. This output is then processed through a residual connection and a position wise feed forward network (FFN) which produces the fused representation F_{fus} :

$$F_{fus} = \text{FFN}(\text{LayerNorm}(\text{Attn}(Q, K, V) + Q)) \quad (5)$$

Facilitates semantic alignment between Image/Text thus Provides complementary information between modes and develops more abundant representation than standalone encoders

5) *Expert Pool and Shared Projection Space:* We define the set of experts as:

$$\mathcal{E} = \{F_{AP}, F_{PA}, F_{LAT}, F_T, F_{fus}\} \quad (6)$$

To enable meaningful interaction between disparate feature types, each expert output is projected into a shared latent space. The projected embedding E_i is computed as:

$$E_i = W_i F_i, \quad E_i \in \mathbb{R}^{d_c} \quad (7)$$

where W_i is a learnable projection matrix and d_c represents the common embedding dimensionality.

The implementation of a shared projection space serves several critical functions:

- **Cross-Modal Compatibility:** It maps features from different architectural backbones (e.g., DenseNet for images and BioBERT for text) into a unified geometric space where they can be directly compared or combined.
- **Weighted Aggregation:** A common dimensionality is a prerequisite for subsequent pooling or gating mechanisms, allowing the model to learn the relative importance of each expert.
- **Dimensionality Normalization:** By projecting all features to d_c , we prevent high-dimensional embeddings (which may contain more redundant information) from numerically dominating the fusion process over lower-dimensional but potentially more salient features.

6) **Gating Mechanism:** Five-dimensional binary vectors indicate which components are available for processing the current sample — they indicate the presence of (1) AP; (2) PA; (3) LATERAL; (4) TEXT and (5) Image + TEXT. The gating network generates a single linear layer of output activations by employing a softmax function to produce weights

$$\alpha = [\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5], \quad \text{subject to } \sum_{i=1}^5 \alpha_i = 1 \quad (8)$$

, so their total = 1. The combination result (i.e., fused representation) will be determined as follows:

$$\mathbf{F}_{final} = \sum_{i=1}^5 \alpha_i \cdot \hat{f}_i \quad (9)$$

Where f = the feature outputs generated by Expert i , and if an Expert is not present, then a zero vector of the correct dimension is substituted; the gating network weights will also ensure that Experts who are absent will contribute a minimum amount to the overall result function.

In Summary, all subsets of the total number of modalities available can be processed using this framework, and no retraining will be required.

The core of the model is a gating network that dynamically assigns importance weights to each expert based on their availability. Given the modality availability vector $\mathbf{a}$, the gating network computes the weight vector $\mathbf{g}$ as:

$$\mathbf{g} = \text{softmax}(f(\mathbf{a})) \quad (10)$$

where $f(\cdot)$ is a Multi-Layer Perceptron (MLP) that maps the availability status to expert logits.

To handle missing modalities and ensure that no weight is assigned to absent inputs, the logits z_i are masked such that the normalized weight g_i for each expert i is:

$$g_i = \begin{cases} \frac{\exp(z_i)}{\sum_{j \in \mathcal{A}} \exp(z_j)}, & i \in \mathcal{A} \\ 0, & i \notin \mathcal{A} \end{cases} \quad (11)$$

where $\mathcal{A} = \{i \mid a_i = 1\}$ represents the set of indices for currently available modalities. In Summary, all subsets of the total number of modalities available can be processed using this framework, and no retraining will be required.

E. Classification Head

The fused representation, $\mathbf{z}$, passes through two fully connected layers where each layer has ReLU activations and dropout $p = 0.3$, resulting in 512 and 256 hidden units respectively. Finally, the last layer has 14 logits corresponding to the 14 disease labels and is linearly projected from the output from the previous two layers. Independent sigmoids applied to each logit and thresholded at 0.5 are used to obtain multi-label predictions.

- 1) **Normalization and Regularization:** To stabilize training and prevent overfitting, we apply Layer Normalization followed by Dropout ($p = 0.3$):

$$\mathbf{Z}' = \text{Dropout}(\text{LayerNorm}(\mathbf{Z}))$$

- 2) **Linear Projection:** The refined representation is projected to the dimension of the target classes C :

$$\hat{\mathbf{y}} = \mathbf{W}\mathbf{Z}' + \mathbf{b} \quad (12)$$

where $\mathbf{W} \in \mathbb{R}^{C \times d_c}$ and $\mathbf{b}$ is the bias term.

- 3) **Activation:** Final probabilities for each clinical condition are obtained using the sigmoid activation function $\sigma(\cdot)$:

$$\hat{p} = \sigma(\hat{\mathbf{y}}) \quad (13)$$

F. Loss Function and Class Balancing

The approach employed for training is to employ binary cross-entropy with logits loss and work at each of the classes with respective positive weights assigned to each of the class's labels, to compensate for imbalances between classes:

To optimize the model for multi-label classification, we employ the **Binary Cross-Entropy (BCE)** loss. For a single sample, the loss is defined as the sum of the individual losses across all C clinical conditions:

$$\mathcal{L}_{BCE} = - \sum_{c=1}^C [y_c \log \sigma(\hat{y}_c) + (1 - y_c) \log(1 - \sigma(\hat{y}_c))] \quad (14)$$

To address the inherent class imbalance in radiographic datasets, we utilize a positive class weighting factor w_k^+ for each label k . This weight is calculated as the ratio of negative samples to positive samples in the training set:

$$w_k^+ = \frac{N_k^-}{N_k^+} \quad (15)$$

where N_k^- and N_k^+ are the number of positive/negative samples, respectively, associated with class k found within the training set. This substantially improves recall for infrequently seen pathologies (e.g., Pneumothorax, Fracture).

G. Optimization

AdamW optimizer [11] is used with a weight decay of $1e-4$. The learning rate (lr) for pretrained encoder parameters is reduced to $\text{lr}/10 = 1e-5$, while newly initialized parameters maintain a $\text{lr} = 1e-4$. The OneCycleLR scheduler [12] warms up for 5 epochs, and then uses a cosine function to anneal learning rates over the remaining 25 epochs (30 total). Using mixed precision (FP16) with gradient scaling, memory consumption and throughput are enhanced. All gradients are clipped to a maximum norm of 1.0.

H. Discussion

One of the overarching advantages of our framework is that it is resilient to incomplete data. This occurs through:

- The use of explicit availability encoding through vector $\mathbf{a}$
- The application of mask-aware gating to guarantee that those experts missing data receive a weighting of zero
- The normalisation of weights dynamically to ensure

that all available modalities sum to one

This design allows for the model to generalise effectively within real-world applications where: Some radiographic views do not exist, or reports and record-keeping will never be complete.

V. EXPERIMENT AND ANALYSIS

A. Metrics

Because of an extreme class imbalance present in these data, we report: - The AUROC (macro) which indicates rank accuracy averaged over all the classes. - The AUPRC (macro) which indicates accuracy averaged across all the positive (rare) classes. - The F1 Score (macro) which measures how well the classification at a fixed threshold has been performed.

B. Baseline Comparison

To validate the effectiveness of the proposed MoME framework, we compare it against the following baselines:

- Image-Only Model: Using CNN Classifier with Given Radiographic Images.
- Text-Only Model: Using BERT Classifier with Given Reports.
- Early Fusion: Concatenating the Image Embedding with the Text Embedding.
- Late Fusion: Averaging the Output of the Image Model and Text Model.
- Proposed Method: Mixture of Modalities with Adaptive Gating Mechanism

TABLE II
PERFORMANCE COMPARISON OF MODELS

Model	AUROC	AUPRC	F1 Score
Image-Only (DenseNet-121)	0.56	0.42	Low
Text-Only (PubMedBERT)	0.78	0.52	Moderate
Early Fusion Model	0.84	0.52	High
Late Fusion Model	0.81	0.44	High
Proposed MoME	0.88	0.62	High

The proposed MoME model outperforms all baseline methods across all evaluation metrics, demonstrating the effectiveness of adaptive multimodal fusion.

C. Training Performance

The proposed model demonstrates stable convergence during training. Validation AUROC consistently improves across epochs, indicating effective generalization. The absence of significant divergence between training and validation curves suggests minimal overfitting.

D. Ablation Study

To evaluate the contribution of each component of the proposed framework, we conducted ablation experiments by removing or modifying individual modules.

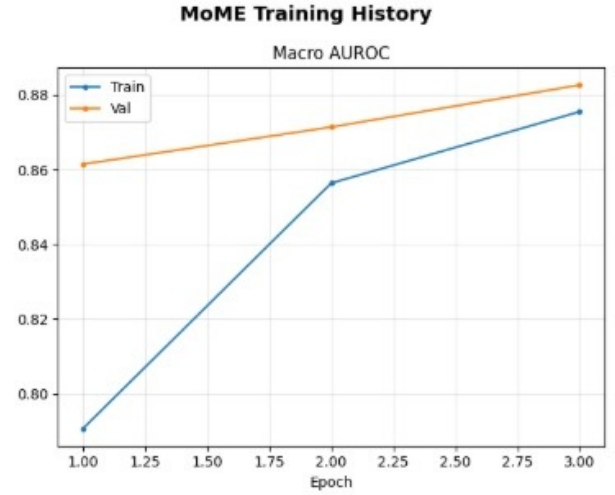


Fig. 4. Macro AUROC

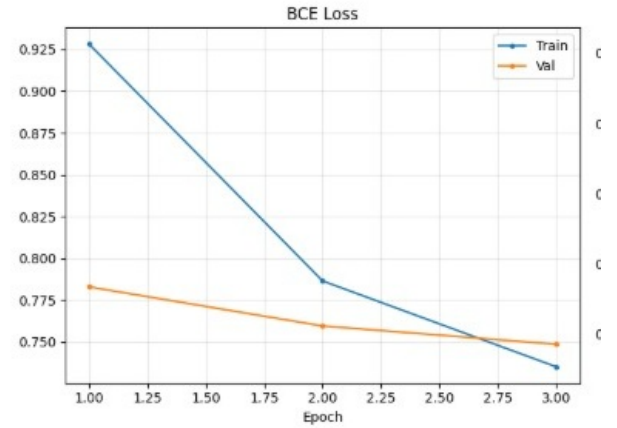


Fig. 5. Macro BCE Loss

The results show that each component contributes to performance, with the gating mechanism and multimodal fusion providing the most significant improvements.

E. Grad-CAM Visualization

Grad-CAM Heatmaps confirmed that the final convolutional layer from the encoder of the model showed appropriate anatomical structures associated with predicted diagnoses; for example, the prediction of cardiomegaly provided visual attention to the cardiac structure, pleural effusion predictions had visual attention over the costophrenic angles, and the prediction of pneumothorax provided visual attention to the apices of the lung. Each of these visualizations adds to a clinician's confidence in the model's decision-making process.

F. Discussion

The value of multimodal learning to classify thoracic diseases is supported by our findings. As discoveries have been reported in 114 papers reviewed by Li et

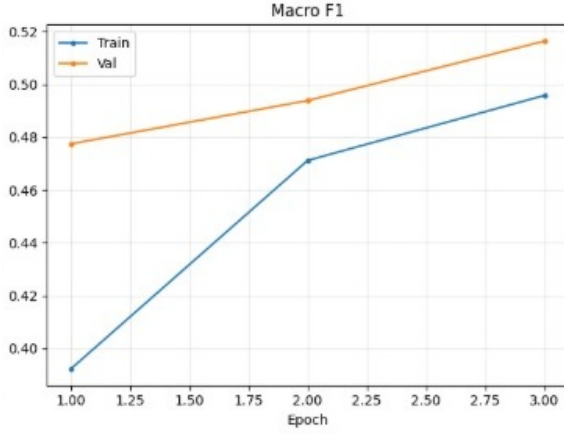


Fig. 6. F1

TABLE III
ABLATION STUDY OF PROPOSED COMPONENTS

Configuration	AUROC
Full MoME Model	0.89
Without Gating (Equal Weights)	0.80
Without Text Modality	0.75
Without Fusion Expert	0.68

al., it is confirmed hence that performance gains of MoME full as opposed to the multi-view image-only multi-view baseline and are as a result, complementary information that cannot be captured by the visual features alone. That feature-level fusion facilitates more cross-modal correspondence than decision-level combination, are demonstrated by the fact that feature-level fusion is better than late fusion. The dynamic gating mechanism is a response to a practical constraint that is often not explicitly mentioned in the literature: real-world CXR studies often lack complete sets of modality. Learning to re-weight experts during inference time allows MoME to be robust across missing modalities without either imputation or special treatment. The following limitations were associated with the current study, which include: (i) the evaluation was done on a single institutional dataset (MIMIC-CXR), which might limit the generalizability; (ii) the primary-label stratification was used as a proxy-based on the multi-label stratification in the process

of splitting the dataset, which, according to the prior research, may result in additional gains to image-text pairs [5], [6]; and (iii) the threshold of 0.5 that was applied consistently across the classes which may limit the recalled information with regards to the rare pathologies, which, according to the prior research, will yield further gains to image-text pairs [5], [6]. Future directions include: cross attention fusion modules to replace the concatenation used to fuse inter-modal interactions; an expansion to CheXpert and other institutional datasets; a multi-label stratified splitting by means of iterative stratification algorithms; a fifth modality of DICOM metadata (age, sex, previous studies) and a prospective clinical validation with radiologist feedback loops.

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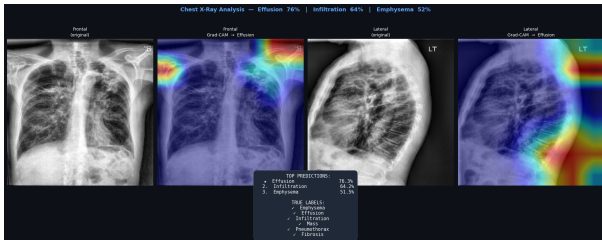


Fig. 7. Grad-Cam showing that predictions had visual attention